

Package ‘BitBreedingSim’

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Author Minoru Inamori

Maintainer Minoru Inamori <inamori@ut-biomet.org>

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add_pop_to_VCF	<i>Add a Population object to a VCF object</i>
----------------	--

Description

This function adds a Population object to an existing VCF object and creates a new VCF object that includes the population information.

Usage

```
add_pop_to_VCF(vcf, pop)
```

Arguments

- | | |
|-----|---|
| vcf | A VCF object. This should be an object created using the relevant VCF functions in the package. |
| pop | A Population object. This should be an object created using the relevant Population functions in the package. |

Value

A new VCF object that includes the added Population information.

Examples

```
# Assume 'vcf' is a valid VCF object and 'pop' is a valid Population object
new_vcf <- add_pop_to_VCF(vcf, pop)
summary(new_vcf) # Inspect the new VCF object
```

add_trait_A	<i>Add Trait with Additive Effects to BaseInfo</i>
-------------	--

Description

This function adds a trait with additive effects to the BaseInfo object.

Usage

```
add_trait_A(
  info,
  name,
  mean,
  h2,
  sd = NULL,
  a = NULL,
  loci = NULL,
  num_loci = 1
)
```

Arguments

info	External pointer to BaseInfo object
name	Name of the trait
mean	Phenotype mean
h2	Heritability
sd	Optional. Phenotype standard deviation
a	Optional. Numeric vector of additive effects
loci	Optional. List of loci in the form of a data frame with two columns: 'chrom' and 'marker'. For example: <pre>chrom <- c(3, 5, 10) marker <- c(1, 2, 1000) loci <- data.frame(chrom, marker)</pre>
num_loci	Optional. Number of loci (default is 1)

Examples

```
# Assume 'info' is a valid BaseInfo object
add_trait_A(info, "Trait1", mean=100.0, h2=0.6, sd=10.0, num_loci = 2)
trait <- get_trait(info, 1)
summary(trait)
```

add_trait_AD

Add Trait with Additive and Dominance Effects to BaseInfo

Description

This function adds a trait with additive and dominance effects to the BaseInfo object.

Usage

```
add_trait_AD(
  info,
  name,
  mean,
  sd = NULL,
  h2 = NULL,
  H2 = NULL,
  a = NULL,
  d = NULL,
  loci = NULL,
  num_loci = 1
)
```

Arguments

info	External pointer to BaseInfo object
name	Name of the trait
mean	Phenotype mean
sd	Optional. Phenotype standard deviation
h2	Optional. Narrow-sense heritability (proportion of variance due to additive genetic effects)
H2	Optional. Broad-sense heritability (proportion of variance due to all genetic effects, can be NULL)
a	Optional. Numeric vector of additive effects
d	Optional. Numeric vector of dominance effects
loci	Optional. List of loci in the form of a data frame with two columns: 'chrom' and 'marker'. For example: <pre>chrom <- c(3, 5, 10) marker <- c(1, 2, 1000) loci <- data.frame(chrom, marker)</pre>
num_loci	Optional. Number of loci (default is 1)

Examples

```
# Assume 'info' is a valid BaseInfo object
add_trait_AD(info, "Trait1", mean=100.0, h2=0.6, H2=0.7, sd=10.0, num_loci = 2)
trait <- get_trait(info, 1)
summary(trait)
```

check_dataframe	<i>Validate the structure and content of the data frame used for crossing information</i>
-----------------	---

Description

This function checks if the input data frame `df` meets the required structure for the `cross_by_table` function. It ensures that `df` is a valid `data.frame`, contains the required columns (`mat`, `pat`, `num`), and verifies that the `num` column has positive numeric values. If any of these checks fail, the function stops execution and displays an error message.

Usage

```
check_dataframe(df)
```

Arguments

<code>df</code>	A data frame. Must include the following columns: - <code>mat</code> : Names of the maternal parents for each cross. - <code>pat</code> : Names of the paternal parents for each cross. - <code>num</code> : Number of offspring to generate for each cross. Must contain positive numeric values.
-----------------	--

Value

No return value. If the validation is successful, the function proceeds silently. If the validation fails, the function stops execution and displays an error message.

Examples

```
# Valid input example
df <- data.frame(mat = c("mat1", "mat2"), pat = c("pat1", "pat2"), num = c(1, 2))
check_dataframe(df) # Passes silently

# Invalid input example: Missing 'num' column
df_invalid <- data.frame(mat = c("mat1", "mat2"), pat = c("pat1", "pat2"))
check_dataframe(df_invalid) # Throws an error
```

 check_parent_existence

Check Parent Existence in Population

Description

This function checks whether the maternal and paternal names in the given cross table are present in the specified maternal and paternal populations.

Usage

```
check_parent_existence(df, mat_pop, pat_pop)
```

Arguments

df	A data.frame representing the cross table containing 'mat' (maternal names) and 'pat' (paternal names) columns.
mat_pop	An external pointer to the maternal Population object.
pat_pop	An external pointer to the paternal Population object.

Value

This function does not return a value. It outputs messages if any maternal or paternal names in the cross table are not found in the respective populations.

Examples

```
# Assuming 'df', 'mat_pop', and 'pat_pop' are valid objects
check_parent_existence(df, mat_pop, pat_pop)
```

 create_base_info

Create a BaseInfo object

Description

This function creates a BaseInfo object. If the seed is set to -1, a random seed is generated, resulting in different outcomes each time the function is called. If a specific seed is provided, the random number generation will be based on that seed, ensuring reproducible results.

Usage

```
create_base_info(
  positions = NULL,
  num_markers = 1000,
  bp = 1e+08,
  chrom_maps = NULL,
  num_chroms = 10,
  cM = 100,
  seed = -1
)
```

Arguments

positions	Optional. A list of numeric vectors, where each vector represents the marker positions for a chromosome in base pairs. The number of elements in positions defines the number of chromosomes and overrides num_chroms. If positions is provided, num_markers is ignored.
num_markers	Optional. An integer. Number of markers per chromosome. Ignored if positions is provided. Default is 1000.
bp	Optional. An integer. Length of each chromosome in base pairs. Ignored if positions is provided. Default is 100000000.
chrom_maps	Optional. A list of data.frames, each representing a chromosome map. Each data.frame should have two columns: 'cM' for centiMorgans and 'position' for base pair positions. The list should be named, with each name corresponding to a chromosome identifier (e.g., "chr1", "chr2", etc.). For a given cM value, the corresponding position can be interpolated or extrapolated, and vice versa. This effectively represents the mapping as a piecewise linear approximation of the chromosome's relationship between cM and base pair positions. If chrom_maps is provided, the parameter num_chroms is ignored.
num_chroms	Optional. An integer. Number of chromosomes. Ignored if either chrom_maps or positions is provided. Default is 10.
cM	Optional. A numeric. Length of each chromosome in centiMorgans. Ignored if chrom_maps is provided. Default is 100.
seed	Optional. An integer. A seed for random number generation. Default is -1, which generates a random seed.

Value

An external pointer to a BaseInfo object.

Examples

```
# Create a BaseInfo object with a random seed
base_info_random <- create_base_info()
get_info(base_info_random)

# Create a BaseInfo object with a specific seed for reproducible results
```

```

base_info_reproducible <- create_base_info(seed = 123)
get_info(base_info_reproducible)

# Create marker positions manually
num_chr <- 10
position <- sapply(1:100, function(i) i * 1000000)
positions <- replicate(10, position, simplify = FALSE)

# Create a chromosome map with 100 cM and 100 Mbp
cM <- c(14.0, 27.9, 40.2, 48.3, 50.0, 51.7, 59.8, 72.1, 86.0, 100.0)
chr_position <- sapply(1:10, function(i) i * 10000000)
chrom_map <- data.frame(cM, chr_position)
chrom_maps <- replicate(10, chrom_map, simplify = FALSE)
names(chrom_maps) <- paste0("chr", 1:10)
info <- create_base_info(positions = positions, chrom_maps = chrom_maps, seed = 123)
get_info(info)

```

```
create_HaploArray_from_pop
```

Create a 3-dimensional HaploArray from a Population object

Description

This function converts a Population object into a 3-dimensional array (HaploArray). The resulting HaploArray contains dimensions representing individuals, markers, and alleles (Maternal and Paternal). The HaploArray has the following dimensions:

dim1 Number of individuals.

dim2 Number of markers.

dim3 Size is 2, representing Maternal and Paternal alleles.

The array also includes dimension names such as individual names, marker names, and allele labels (Maternal and Paternal).

Usage

```
create_HaploArray_from_pop(pop)
```

Arguments

pop	An external pointer to a Population object. This object must be properly initialized.
-----	---

Details

Note: This function is the reverse of create_pop_from_HaploArray, which creates a Population object from a HaploArray.

Value

A 3-dimensional array (HaploArray) containing the following dimensions:

Individuals Individual names as rows.

Markers Marker names as columns.

Alleles Maternal and Paternal alleles as the third dimension.

Examples

```
info <- create_base_info(num_chroms=2, num_markers=10, cM=100, bp=1e6, seed=2)
add_trait_A(info, "Trait1", mean=100.0, h2=0.6, sd=10.0, num_loci = 2)
trait <- get_trait(info, 1)

haploArray <- array(
  data = rbinom(n = 120, size = 1, prob = 0.3),
  dim = c(3, 20, 2),
  dimnames = list(
    paste0("Ind_", 1:3),
    paste0("Mrk_", 1:20),
    c("Maternal", "Paternal")
  )
)
pop <- create_pop_from_HaploArray(haploArray, info)
prog <- cross_randomly(pop, pop, num_inds = 5, name_base = "prog_")
haploArrayProg <- create_HaploArray_from_pop(prog)
```

```
create_info_pop_from_VCF
```

Create BaseInfo and Population from a VCF file

Description

This function takes a VCF object and a seed value, and returns both a Population object and its associated BaseInfo object. It reads the input VCF and initializes the data accordingly. The seed value is used to initialize the pseudo-random number generator for the BaseInfo object. If the seed is set to -1, an appropriate value will be automatically chosen.

Usage

```
create_info_pop_from_VCF(vcf, seed = -1)
```

Arguments

vcf	An external pointer to a VCF object.
seed	An integer. The seed value for initializing the BaseInfo object's pseudo-random number generator. Defaults to -1, which automatically selects a suitable seed.

Value

A list containing two elements:

info An external pointer to a BaseInfo object.

pop An external pointer to a Population object.

Examples

```
# Assuming 'vcf_file' is a valid VCF file
vcf <- read_VCF(vcf_file)
result <- create_info_pop_from_VCF(vcf, seed = 42)
summary(result$info)
summary(result$pop)
```

create_origins

Create origins for a Population object

Description

Create origins for a Population object

Usage

```
create_origins(
  info,
  genotype_ratio = NULL,
  names = NULL,
  num_inds = NULL,
  name_base = NULL
)
```

Arguments

info	An external pointer to a BaseInfo object.
genotype_ratio	A numeric vector of length 3 representing the desired ratio of genotypes (0/0, 0/1, 1/1). All elements must be non-negative, and the vector must not be all zeros. The values will be normalized to sum to 1. Default is <code>c(0.25, 0.5, 0.25)</code> .
names	A character vector. The names of individuals. Either <code>names</code> or <code>name_base</code> must be provided, but not both.
num_inds	An integer. The number of individuals.
name_base	A string. The base name for individuals.

Value

An external pointer to a Population object.

Examples

```
# Example 1: Specify individual names directly
pop <- create_origins(info, names = c("p1", "p2"))
get_individual_names(pop)

# Example 2: Generate individual names using name_base and num_inds
pop2 <- create_origins(info, num_inds = 2, name_base = "q")
get_individual_names(pop2)

# Example 3: Specify genotype_ratio explicitly
pop3 <- create_origins(info, genotype_ratio = c(1, 1, 1), num_inds = 3, name_base = "g")
get_individual_names(pop3)
```

```
create_pop_from_HaploArray
```

Create a Population object from a HaploArray

Description

This function takes a 3-dimensional array (HaploArray) and a BaseInfo object to create a Population object. The HaploArray should have dimensions representing individuals, markers, and alleles (Maternal and Paternal). Allele values must be binary (0 or 1). A value of 0 indicates that the allele matches the reference genome, while a value of 1 indicates the alternative allele. The function validates the input data and initializes the Population object accordingly.

Usage

```
create_pop_from_HaploArray(haploArray, info)
```

Arguments

haploArray	A 3-dimensional array where: dim1 Number of individuals. dim2 Number of markers. dim3 Size must be 2, representing Maternal and Paternal alleles. Allele values must be binary (0 or 1), based on the reference genome.
info	An external pointer to a BaseInfo object.

Value

An external pointer to a Population object.

Examples

```
# Create a HaploArray
haploArray <- array(
  data = rbinom(n = 30, size = 1, prob = 0.3),
  dim = c(3, 5, 2),
  dimnames = list(
    paste0("Ind_", 1:3),
    paste0("Mrk_", 1:5),
    c("Maternal", "Paternal")
  )
)

# Assuming 'info' is a valid BaseInfo object
pop <- create_pop_from_HaploArray(haploArray, info)
summary(pop)
```

cross_by_table	<i>Cross populations according to a table</i>
----------------	---

Description

This function performs crossing of individuals from two Population objects (maternal and paternal parents) based on specified crossing information provided in a table. Each row of the table defines a specific cross, including the number of offspring to generate for that cross. You can specify either name_base or provide a complete names vector for the offspring.

Usage

```
cross_by_table(
  df,
  mat_pop,
  pat_pop,
  names = NULL,
  name_base = NULL,
  num_threads = 0
)
```

Arguments

df	A data frame containing crossing information. It should include the following columns: mat Names of the maternal parents for each cross. pat Names of the paternal parents for each cross. num Number of offspring to generate for each cross.
mat_pop	A Population object representing the maternal parents. This should be an object created using the relevant Population functions in the package.

pat_pop	A Population object representing the paternal parents. This should be an object created using the relevant Population functions in the package.
names	Optional. A character vector containing specific names for individuals. If provided, the length of names should match the total number of offspring specified in df.
name_base	Optional. A character string used as the base name for generating individual names. For example, if name_base = "prog_", the generated names will follow the format c("prog_1", "prog_2", ...).
num_threads	Optional. A positive integer representing the number of threads to use. If not specified or set to 0, the maximum number of available threads is used.

Value

A Population object with offspring individuals.

Examples

```
# Example using name_base
df <- data.frame(mat = c("mat1", "mat2"), pat = c("pat1", "pat2"), num = c(1, 2))
new_population <- cross_by_table(df, mat_pop, pat_pop, name_base = "prog_")

# Example using predefined names
names_vector <- c("child1", "child2", "child3")
new_population <- cross_by_table(df, mat_pop, pat_pop, names = names_vector)

# Summary of the new Population object
summary(new_population)
```

cross_randomly	<i>Cross Population randomly</i>
----------------	----------------------------------

Description

This function performs random crossing of individuals from two Population objects (maternal and paternal parents) to generate a new Population object. For each offspring, a parent is randomly selected from each Population (one from the maternal Population and one from the paternal Population). You can specify either name_base and num_inds or provide a complete names vector for the offspring.

Usage

```
cross_randomly(
  mat_pop,
  pat_pop,
  names = NULL,
  num_inds = NULL,
  name_base = NULL,
```

```

    start_index = 1,
    allow_selfing = TRUE,
    num_threads = 0
  )

```

Arguments

<code>mat_pop</code>	A Population object representing the maternal parents. This should be an object created using the relevant Population functions in the package.
<code>pat_pop</code>	A Population object representing the paternal parents. This should be an object created using the relevant Population functions in the package.
<code>names</code>	Optional. A character vector containing specific names for individuals. If provided, the length of names determines the number of offspring.
<code>num_inds</code>	Optional. A positive integer representing the number of offspring. Required if <code>name_base</code> is specified.
<code>name_base</code>	Optional. A character string used as the base name for generating individual names. For example, if <code>name_base = "p"</code> and <code>num_inds = 3</code> , the generated names will be <code>c("p1", "p2", "p3")</code> . This parameter must be used in combination with <code>num_inds</code> .
<code>start_index</code>	Optional. A positive integer specifying the starting index used when generating offspring names from <code>name_base</code> . For example, if <code>name_base = "child_"</code> , <code>num_inds = 3</code> , and <code>start_index = 5</code> , the generated names will be <code>c("child_5", "child_6", "child_7")</code> . This parameter is ignored when <code>names</code> is provided.
<code>allow_selfing</code>	Logical. If <code>FALSE</code> , selfing (crossing between individuals with the same name) is not allowed. Defaults to <code>TRUE</code> .
<code>num_threads</code>	Optional. A positive integer representing the number of threads to use. If not specified or set to 0, the maximum number of available threads is used.

Details

If `allow_selfing = FALSE`, the function prevents selfing (i.e., crossing between two individuals with the same name). If both populations contain only one individual with the same name, the function will raise an error.

Value

A Population object with offspring individuals.

Examples

```

# Example using name_base and num_inds
new_population <- cross_randomly(mothers, fathers, num_inds = 100, name_base = "offspring_")

# Example using predefined names
names_vector <- c("child1", "child2", "child3")
new_population <- cross_randomly(mothers, fathers, names = names_vector)

# Disallow selfing

```

```
new_population <- cross_randomly(mothers, fathers, num_inds = 10, name_base = "child_", allow_selfing = FALSE)

# Summary of the new Population object
summary(new_population)
```

extract_map_data	<i>Extract chromosome and position data from a Map object</i>
------------------	---

Description

This function retrieves detailed data from a Map object, including chromosome names and marker positions, and returns the data as a data.frame for easier manipulation.

Usage

```
extract_map_data(map)
```

Arguments

map	An external pointer to a Map object. This object must be properly initialized and created using appropriate functions.
-----	--

Value

A data.frame containing the following columns:

chrom A character vector containing the names of chromosomes.

position An integer vector specifying the positions of markers along the chromosomes.

Examples

```
# Assuming 'map' is a valid Map object
map_data <- extract_map_data(map)
head(map_data) # Display the first few rows of the data.frame
```

generate_names	<i>Generate names for individuals based on a base name and number of individuals</i>
----------------	--

Description

This function creates a vector of names for individuals when names is not provided. It uses a base name (name_base) and a specified number of individuals (num_inds) to generate names in the format name_base1, name_base2, ..., up to the total number.

Usage

```
generate_names(name_base, num_inds, start_index = 1)
```

Arguments

name_base A character string used as the base for generating names. For example, `name_base = "p"` will generate names like "p1", "p2",

num_inds A positive integer indicating the number of names to generate.

start_index Optional. A positive integer specifying the starting index for numbering. For example, `start_index = 5` will generate names like "p5", "p6",

Value

A character vector of generated names.

Examples

```
# Generate 3 names with base "p"
generated_names <- generate_names("p", 3)
print(generated_names) # Output: "p1", "p2", "p3"
```

`get_genotypic_values` *Get genotypic values from a Population object*

Description

Get genotypic values from a Population object

Usage

```
get_genotypic_values(pop, i)
```

Arguments

pop An external pointer to a Population object.

i An integer index representing the trait for which genotypic values are to be retrieved. The index should be between 1 and the total number of traits available in the Population object.

Value

A vector of genotypic values.

Examples

```
# Assuming 'pop' is a valid Population object and trait index 1 is valid
genotypic_values <- get_genotypic_values(pop, 1)
print(genotypic_values)
```

get_genotypes	<i>Get genotypes from a Population object</i>
---------------	---

Description

This function retrieves the genotypes from a given Population object. The genotypes are represented in a matrix format where rows correspond to individuals and columns correspond to markers.

Usage

```
get_genotypes(pop)
```

Arguments

pop An external pointer to a Population object.

Details

Genotype encoding:

- 0/0 is encoded as -1
- 0/1 is encoded as 0
- 1/1 is encoded as 1

Value

A matrix of genotypes where rows are individuals and columns are markers.

Examples

```
info <- create_base_info()
pop <- create_origins(info, num_inds = 2, names = c("p1", "p2"))
geno <- get_genotypes(pop)
geno[, 1:5]
```

get_individual_names	<i>Get name data from a Population object</i>
----------------------	---

Description

Get name data from a Population object

Usage

```
get_individual_names(pop)
```

Arguments

pop	An external pointer to a Population object.
-----	---

Value

A data.frame containing the following columns:

name Names from the Population object

mat Maternal names from the Population object

pat Paternal names from the Population object

Examples

```
# Assuming 'pop' is a valid Population object
name_data <- get_individual_names(pop)
print(name_data)
```

get_info	<i>Retrieve key values from a BaseInfo object</i>
----------	---

Description

This function extracts essential information from a BaseInfo object. A BaseInfo object stores meta-data related to genetic simulations, such as the number of chromosomes, traits, and markers. This function is useful for summarizing or verifying the content of a BaseInfo object during analysis.

Usage

```
get_info(info)
```

Arguments

info	An external pointer to a BaseInfo object. This object must be properly initialized and created using the appropriate BaseInfo constructor or relevant functions.
------	--

Value

A list containing the following elements:

num_chroms An integer representing the total number of chromosomes contained in the BaseInfo object.

num_traits An integer indicating the number of traits managed by the BaseInfo object.

num_markers An integer specifying the total number of markers tracked in the BaseInfo object.

Examples

```
# Assume 'info' is a valid BaseInfo object
values <- get_info(info)
print(values$num_chroms) # Print the number of chromosomes
print(values$num_traits) # Print the number of traits
print(values$num_markers) # Print the number of markers
```

get_map

Retrieve Genetic Map Information

Description

This function retrieves the genetic map information from a BaseInfo object.

Usage

```
get_map(info)
```

Arguments

info An object of class BaseInfo. This object contains the genetic map information.

Value

A list of data frames, each representing a chromosome. Each data frame contains two columns:

- **cM**: The centiMorgan positions of the markers.
- **position**: The base pair positions of the markers.

Examples

```
## Not run:
# Assuming `info` is a valid BaseInfo object
map <- get_map(info)
print(map)

## End(Not run)
```

get_phased_genotypes *Get phased genotypes from a Population object*

Description

This function retrieves the genotypes from a given Population object. The genotypes are represented in a matrix format where rows correspond to markers and columns correspond to individuals.

Usage

```
get_phased_genotypes(pop)
```

Arguments

pop An external pointer to a Population object.

Details

Genotype is 0|0, 0|1, 1|0, or 1|1

Value

A matrix of genotypes where rows are individuals and columns are markers.

Examples

```
info <- create_base_info()
pop <- create_origins(info, num_inds = 2, names = c("p1", "p2"))
geno <- get_phased_genotypes(pop)
geno[, 1:5]
```

get_phased_int_genotypes *Get phased integer genotypes from a Population object*

Description

This function retrieves the genotypes from a given Population object. The genotypes are represented in a matrix format where rows correspond to individuals and columns correspond to markers. Each individual has two rows: the first row represents the maternal allele and the second row represents the paternal allele.

Usage

```
get_phased_int_genotypes(pop)
```

Arguments

pop An external pointer to a Population object.

Details

Genotype is represented as integers: 0 or 1

Value

A matrix of genotypes where rows are individuals and columns are markers. Each individual has two rows: the first row is the maternal allele and the second row is the paternal allele.

Examples

```
info <- create_base_info()
pop <- create_origins(info, num_inds = 2, names = c("p1", "p2"))
geno <- get_phased_int_genotypes(pop)
geno[, 1:5]
```

get_phenotypes	<i>Get phenotypes from a Population object</i>
----------------	--

Description

Get phenotypes from a Population object

Usage

```
get_phenotypes(pop, i)
```

Arguments

pop An external pointer to a Population object.

i An integer index representing the trait for which phenotypes are to be retrieved. The index should be between 1 and the total number of traits available in the Population object.

Value

A vector of phenotypes.

Examples

```
# Assuming 'pop' is a valid Population object and trait index 1 is valid
phenotypes <- get_phenotypes(pop, 1)
print(phenotypes)
```

get_pop_info	<i>Get information for a Population object</i>
--------------	--

Description

This function retrieves detailed information about a Population object, including the number of individuals, the number of chromosomes, and the number of markers in the population.

Usage

```
get_pop_info(pop)
```

Arguments

pop	A Population object. This should be an object created using the relevant Population functions in the package.
-----	---

Value

A list containing: - num_individuals: The number of individuals in the population. - num_chromosomes: The number of chromosomes in the population. - num_markers: The number of markers in the population.

Examples

```
# Create base information
info <- create_base_info()
# Create a population with 2 individuals
pop <- create_origins(info, num_inds = 2, names = c("p1", "p2"))
# Retrieve population information
get_pop_info(pop)
```

get_trait	<i>Get a trait from a BaseInfo object</i>
-----------	---

Description

Get a trait from a BaseInfo object

Usage

```
get_trait(info, i)
```

Arguments

info	An external pointer to a BaseInfo object.
i	An integer. The index of the trait to retrieve.

Value

A Trait object containing the following elements:

name The name of the trait as a character string.

type The type of the trait, such as "Additive Effect Only".

mean The mean value of the trait as a numeric.

sd The standard deviation of the trait as a numeric.

h2 The narrow-sense heritability of the trait as a numeric (between 0 and 1).

H2 (Optional) The broad-sense heritability of the trait as a numeric.

hasdominants A logical value indicating whether the trait includes dominant effects.

loci A list of loci associated with the trait. Each element is a pair containing detailed information about a locus.

additives The additive effects for each locus as a numeric vector.

dominants The dominant effects for each locus as a numeric vector (if applicable).

Examples

```
# Assume 'info' is a valid BaseInfo object
trait <- get_trait(info, 1)
summary(trait)

# Access specific trait properties
print(trait$name) # Print the name of the trait
print(trait$loci) # Print the list of loci
```

get_VCF_info

Get information from a VCF object

Description

This function retrieves detailed information about a VCF object, including the number of individuals and the number of markers contained within the VCF.

Usage

```
get_VCF_info(vcf)
```

Arguments

vcf A VCF object. This should be an object created using the relevant VCF functions in the package.

Value

A list containing: - num_individuals: The number of individuals in the VCF object. - num_markers: The number of markers in the VCF object.

Examples

```
# Assume 'vcf' is a valid VCF object
vcf_info <- get_VCF_info(vcf)
print(vcf_info) # Displays the number of individuals and markers
```

join_pops

Join multiple Population objects

Description

This function takes two or more Population objects and merges them into a single Population. Inputs may be passed as separate arguments, as a vector (e.g., `c(pop1, pop2)`), or as a list.

Usage

```
join_pops(...)
```

Arguments

... Population objects or collections (list/vector) of Population objects.

Details

Inputs are flattened so that the following forms are equivalent:

- `join_pops(pop1, pop2, pop3)`
- `join_pops(c(pop1, pop2, pop3))`
- `join_pops(list(pop1, pop2, pop3))`
- `do.call(join_pops, list(pop1, pop2, pop3))`

All elements are checked to ensure they inherit from the "Population" class.

Value

A new Population object containing all individuals from the provided populations.

Examples

```
# Example 1: Basic usage
# Assuming 'pop1', 'pop2', and 'pop3' are valid Population objects
combined_pop <- join_pops(pop1, pop2, pop3)
print(combined_pop)

# Example 2: Using c() to combine inputs
combined_pop2 <- join_pops(c(pop1, pop2, pop3))

# Example 3: Using a list
pops_list <- list(pop1, pop2, pop3)
combined_pop3 <- join_pops(pops_list)
```

 modify_trait_parameters

Modify Trait Parameters

Description

This function allows modifying the parameters of a specified trait in a BaseInfo object. At least one of h2, H2, a, additive_multiplier, d, or dominant_multiplier must be provided. The function ensures that only valid parameter combinations are specified and updates the heritability, additive effects, and/or dominant effects accordingly.

Usage

```
modify_trait_parameters(
  info,
  i,
  h2 = NULL,
  H2 = NULL,
  a = NULL,
  additive_multiplier = NULL,
  d = NULL,
  dominant_multiplier = NULL
)
```

Arguments

info	A BaseInfo object containing the trait information.
i	An integer index of the trait to be modified. Must be a positive integer within the valid range.
h2	(Optional) Narrow-sense heritability. If specified, it adjusts the variance components according to the provided value.
H2	(Optional) Broad-sense heritability. Can be specified alongside h2, and may also be combined with a or additive_multiplier. However, it cannot be specified with d or dominant_multiplier.
a	(Optional) A numeric vector specifying the additive effects for each QTL. If specified alongside h2, the total variance (additive, dominant, and error variance) is updated.
additive_multiplier	(Optional) A scalar to scale the current additive effects. If specified alongside h2, the total variance is updated while keeping the additive effects scaled.
d	(Optional) A numeric vector specifying the dominant effects for each QTL. Cannot be specified with H2.
dominant_multiplier	(Optional) A scalar to scale the current dominant effects. Cannot be specified with H2.

Notes:

- Only one of a or additive_multiplier can be specified at the same time.
- Similarly, only one of d or dominant_multiplier can be specified at the same time.
- H2, d, and dominant_multiplier are mutually exclusive; at most one of these parameters can be provided at a time.

Behavior:

- If only h2 is specified, it adjusts the error variance to match the specified h2. Additionally, the additive effects are scaled proportionally (vector multiplied by a constant) to achieve the specified h2, while the dominant effects remain unchanged.
- If a or additive_multiplier is specified alongside h2, the total variance (additive, dominant, and error variance) is updated accordingly.
- If d or dominant_multiplier is specified alongside h2, the total variance remains unchanged, and only dominant effects are updated.
- If only a, additive_multiplier, d, or dominant_multiplier is specified, the respective effects are updated, and heritability is adjusted accordingly.

Value

No return value. The function modifies the parameters of the specified trait directly.

Examples

```
info <- create_base_info()
add_trait_AD(info, "Trait1", mean = 100.0, h2 = 0.6, H2 = 0.7, sd = 10.0, num_loci = 2)

# Modify narrow-sense heritability only
modify_trait_parameters(info, 1, h2 = 0.7)

# Modify broad-sense heritability only
modify_trait_parameters(info, 1, H2 = 0.8)

# Modify heritability and additive effects
modify_trait_parameters(info, 1, h2 = 0.7, a = c(11, 12))

# Modify heritability and dominant effects
modify_trait_parameters(info, 1, h2 = 0.7, d = c(3, 4))

# Modify heritability and scale additive effects
modify_trait_parameters(info, 1, h2 = 0.7, additive_multiplier = 1.1)

# Modify heritability and scale dominant effects
modify_trait_parameters(info, 1, h2 = 0.7, dominant_multiplier = 1.2)

# Modify additive effects only
modify_trait_parameters(info, 1, a = c(11, 12))

# Scale dominant effects only
modify_trait_parameters(info, 1, dominant_multiplier = 1.2)
```

```
# Modify heritability, additive effects, and dominant effects together
modify_trait_parameters(info, 1, h2 = 0.75, a = c(10, 12), d = c(5, 6))
```

print_trait

Print detailed information about a Trait object

Description

This function prints the detailed characteristics of a Trait object in a human-readable format. A Trait object represents a characteristic or feature in genetic simulations, and this function displays its metadata such as its name, type, mean, standard deviation, heritability (h2), associated loci, additive effects, and dominant effects (if applicable).

Usage

```
print_trait(trait, ...)
```

Arguments

trait	A Trait object containing the following elements: name The name of the trait as a character string. type The type of the trait (e.g., "quantitative" or "binary"). mean The mean value of the trait as a numeric. sd The standard deviation of the trait as a numeric. h2 The narrow-sense heritability of the trait as a numeric (between 0 and 1). H2 (Optional) The broad-sense heritability of the trait as a numeric (if applicable). hasdominants A logical value indicating whether the trait includes dominant effects. loci A list of loci associated with the trait, formatted as coordinate pairs. additives A numeric vector representing the additive effects of the trait. dominants (Optional) A numeric vector representing the dominant effects of the trait (if applicable).
...	Additional arguments (not currently used).

Details

The loci are formatted as coordinate pairs, representing their positions. Additive and dominant effects are printed as space-separated values for clarity.

Value

Prints the detailed information of the Trait object directly to the console.

Examples

```
info <- create_base_info(num_chroms=2, num_markers=10, cM=100, bp=1e6, seed=2)
addTraitA(info, "Trait1", mean=100.0, h2=0.6, sd=10.0, num_loci = 2)
trait <- getTrait(info, 1)
print_trait(trait)

# Alternatively, use the generic print() function
print(trait)
```

read_VCF

Read a VCF file and return a VCF object

Description

This function reads a VCF file from the specified filename and returns a VCF object as an external pointer. If the input is not a valid character string, a message is displayed and NULL is returned.

Usage

```
read_VCF(filename)
```

Arguments

filename A character string specifying the path to the VCF file.

Value

An external pointer to a VCF object, or NULL if an error occurs.

Examples

```
# Assuming 'example.vcf' is a valid VCF file
vcf <- read_VCF("example.vcf")
if (is.null(vcf)) {
  cat("Failed to read the VCF file.\n")
}
```

remove_pop	<i>Remove individuals from a Population object</i>
------------	--

Description

Remove individuals from a Population object

Usage

```
remove_pop(pop, samples)
```

Arguments

pop	An external pointer to a Population object.
samples	A vector of either integer indices or character names representing the individuals to be removed.

Value

An external pointer to a new Population object with the specified individuals removed.

Examples

```
# Assuming 'pop' is a valid Population object
# Remove individuals by index
new_pop <- remove_pop(pop, c(1, 2, 3))
get_individual_names(new_pop)

# Remove individuals by name
new_pop2 <- remove_pop(pop, c("ind1", "ind2"))
get_individual_names(new_pop2)
```

select_pop	<i>Select individuals from a Population object</i>
------------	--

Description

Select individuals from a Population object

Usage

```
select_pop(pop, indices)
```

Arguments

pop	An external pointer to a Population object.
indices	A vector of integer indices representing the individuals to be selected.

Value

An external pointer to a new Population object containing the selected individuals.

Examples

```
# Assuming 'pop' is a valid Population object and indices are valid
selected_pop <- select_pop(pop, c(1, 2, 3))
get_individual_names(selected_pop)
```

set_individual_names *Set individual names for a Population object*

Description

This function assigns names to individuals in a Population object. The names argument must be a character vector, and pop must be an object of the "Population" class.

Usage

```
set_individual_names(names, pop)
```

Arguments

names	A character vector containing the names of individuals to assign.
pop	A Population object where the individual names will be set.

Value

None. The function modifies the Population object directly.

Examples

```
# Create a Population object and set individual names
info <- create_base_info()
pop <- create_origins(info, num_inds = 2, name_base = "ind")
set_individual_names(c("sample1", "sample2"), pop)

# Access the updated individual names
get_individual_names(pop)
```

summary_map	<i>Summarize the details of a Map object</i>
-------------	--

Description

This function extracts and displays key information from a Map object. A Map object represents the genetic map used in simulations, containing details about chromosomes and points (e.g., markers or loci) along the map. The summary provides an overview of the structure and size of the map.

Usage

```
summary_map(map, ...)
```

Arguments

map	An external pointer to a Map object. This object must be properly initialized and created using the appropriate functions.
...	Additional arguments (not currently used).

Value

Prints a formatted summary of the Map object directly to the console.

Examples

```
# Assuming 'map' is a valid Map object
summary_map(map)
```

summary_population	<i>Summarize the details of a Population object</i>
--------------------	---

Description

This function extracts and displays key information from a Population object. A Population object represents a collection of individuals, chromosomes, and genetic markers used in genetic simulations. The summary includes the number of individuals, chromosomes, and markers in the population, providing an overview of its structure and size.

Usage

```
summary_population(pop, ...)
```

Arguments

pop	An external pointer to a Population object. This object must be properly initialized and created using the appropriate functions or constructors.
...	Additional arguments (not currently used).

Value

Prints a formatted summary of the Population object directly to the console.

Examples

```
# Assuming 'pop' is a valid Population object
summary_population(pop)
```

summary_trait	<i>Summarize the details of a Trait object</i>
---------------	--

Description

This function provides a formatted summary of a Trait object. A Trait object represents a characteristic or feature in genetic simulations, and contains metadata such as its name, type, mean, standard deviation, heritability (h2), and information about associated loci. If the trait includes dominant effects, the broad-sense heritability (H2) is also displayed.

Usage

```
summary_trait(trait)
```

Arguments

trait	A Trait object containing the following elements:
name	The name of the trait as a character string.
type	The type of the trait (e.g., "Additive Effect Only").
mean	The mean value of the trait as a numeric.
sd	The standard deviation of the trait as a numeric.
h2	The narrow-sense heritability of the trait as a numeric (between 0 and 1).
H2	(Optional) The broad-sense heritability of the trait as a numeric (if applicable).
hasdominants	A logical value indicating whether the trait includes dominant effects.
loci	A list of loci associated with the trait.

Value

Prints a summary of the Trait object directly to the console.

Examples

```
info <- create_base_info(num_chroms=2, num_markers=10, cM=100, bp=1e6, seed=2)
addTraitA(info, "Trait1", mean=100.0, h2=0.6, sd=10.0, num_loci = 2)
trait <- getTrait(info, 1)
summary_trait(trait)

# Using the generic summary() function
summary(trait)
```

write_VCF

Write VCF or Population object to a VCF file

Description

This function writes either a VCF object or a Population object to a VCF file. If the input is a VCF object, it directly writes the VCF data to the specified file. If the input is a Population object, it generates VCF data from the Population and writes it to the file.

Usage

```
write_VCF(obj, filename)
```

Arguments

obj	An object of class VCF or Population. The data to write to the VCF file.
filename	A string specifying the path to the output VCF file.

Examples

```
# Example 1: Write a Population object to a VCF file
pop <- create_origins(info, num_inds = 2, names = c("p1", "p2"))
write_VCF(pop, "population_output.vcf")

# Example 2: Write a VCF object to a VCF file
vcf <- read_VCF("input.vcf")
write_VCF(vcf, "vcf_output.vcf")
```

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